Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

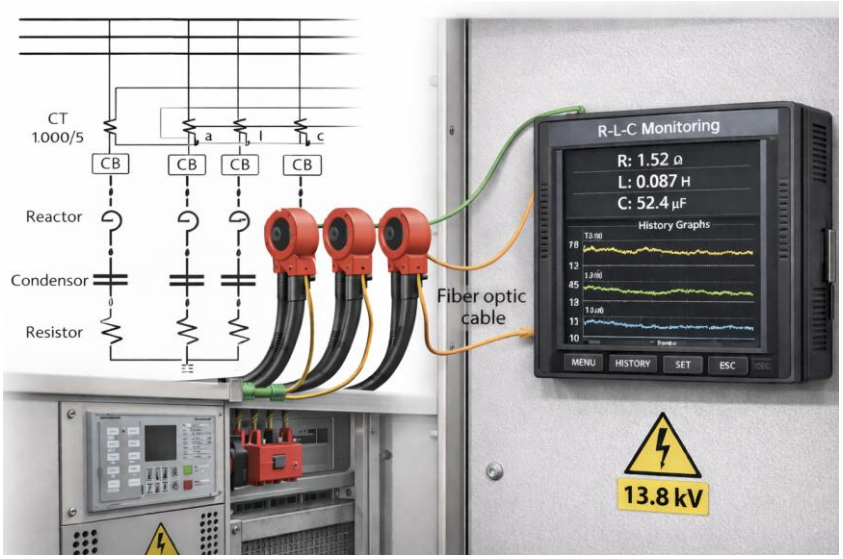
Title of Proposed Technology	**Preventive diagnostic technology for nuclear generator coolant motor utilizing non-invasive online RLC load tracking and time-series-based diagnosis**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power (√), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Energy & Resources*	
	Medium Category	Small Category	Medium Category	Small Category
	Nuclear Power	*Nuclear Plant Operation & Maintenance*	*Transmission & Distribution Systems*	*Power System Monitoring & Operation Technologies*
Concept of Proposed Technology	**Core Concept** The goal of this R&D project is to develop an intelligent system that tracks and diagnoses the real-time condition of large-capacity power equipment (motors, transformers, generators, power reactors, power capacitors) by monitoring and analyzing their equivalent load (RLC) in real time. The load-tracking-based equipment for fault diagnosis to be developed will incorporate artificial intelligence and use learning-based methods to detect subtle changes in the condition of power equipment. If a change indicates a potential fault, the system will diagnose the problem in advance and notify operators, thereby preventing failures. For example, the reactor coolant pump motor in a nuclear power plant is one of the most critical pieces of power equipment; a failure would immediately shut down the entire plant. However, a real-time condition monitoring system for this equipment has yet to be developed. Currently, when a failure occurs, the nuclear power plant shuts down, and the			

equipment is replaced as a post-failure measure. This repeated process is a significant social issue and results in substantial economic losses.

Similarly, for reactive power compensation devices (reactors and capacitors), there is no system for tracking and diagnosing equipment condition. Failures are addressed only after they occur, requiring equipment replacement. Therefore, the introduction of a real-time preventive diagnosis system is necessary.

Key Functions and Potential Applications

The device to be developed in this project extracts real-time R-L-C load values and uses learning-based methods to detect subtle changes in condition, thereby identifying fault symptoms in advance and preventing accidents. This requires highly precise non-invasive sensing and diagnostic technology, including the measurement of very small currents in the microampere range, phase measurement accuracy of 0.05 degrees, high-order harmonic identification algorithms, and the ability to decompose and extract R-L-C load components from sensing data and harmonics. Non-invasive sensing is crucial for retrofitting existing equipment. For high-precision sensing and diagnostics, the project aims to develop a non-invasive (e.g., magnetic-field sensing) digital current-sensing technology tailored to the specific purpose, signal-processing algorithms for harmonic and RLC extraction, and technology for learning the operating conditions of high-voltage equipment using artificial intelligence. The real-time diagnostic technology will be applied to high-voltage equipment, such as nuclear generator coolant motor, with voltages ranging from 480V to 765kV. A wide voltage range is important for versatile applications across a wide range of power-capacity equipment.



Rationale for Joint R&D Project

Critical Technical Barriers & Why Global Collaboration?

This joint research and development project aims to overcome limitations by integrating the following elements: non-contact magnetic sensing technology for high-precision current measurement; real-time R-L-C component extraction via voltage and current harmonic and correlation analysis; and AI-based algorithms for power equipment degradation and lifespan prediction utilizing the real-time monitored data. Collaboration between Purdue University, which has expertise in sensing and diagnostic technologies, and a Korean SME with strong capabilities in ICT-

	based AI algorithms and system integration is essential to accelerate technological maturity and global market entry. In particular, since direct installation on large-capacity power equipment in power generation and transmission facilities is essential for device demonstration and verification, collaboration with a Korean SME that has a deep understanding of power equipment, a history of power equipment testing, and experience in developing magnetic field sensor-based devices is crucial for optimizing and verifying the performance of the developed devices and solutions for large-capacity motors and power compensation devices.
Key Words	**1. R-L-C Extraction** **2. Non-Contact Current Monitoring** **3. AI-Based Load Health Diagnosis** **4. Nuclear Reactor Coolant Pump** **5. Online Monitoring & Diagnosis**

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (7)
R&D Objectives	**R&D Objectives** The objectives of this research and development project are as follows: 1. To develop a minimally intrusive voltage and current sensing and monitoring solution optimized for R-L-C load diagnosis of power equipment. 2. To develop a low-power sensing and data acquisition module that can be retrofit-installed in a non-intrusive manner on existing equipment. 3. To develop an AI-based algorithm for harmonic and R-L-C extraction and condition diagnosis. 4. To implement a wireless remote inspection and monitoring system for preventive maintenance of high-power equipment. 5. To verify the feasibility of complying with electromagnetic regulations and requirements.			
R&D Target Specifications	The target specifications of this research and development project are as follows: • R-L-C extraction - Phase resolution: <0.05 degree - Harmonic support: up to 10th order - RLC extraction accuracy: >98% • Operation temperature range: –20°C ~ +60°C • Equipment voltage range: 480V ~ 765kV • External data links: Ethernet, Wi-Fi, BT			

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	• Non-invasive Current Sensing Specification - Effective Bandwidth: 10Hz – 1kHz - Power: ≤ 50mW - Rated Current: 0 - 30A - Current Accuracy: >99% - Withstand Voltage: 3000V isolation. - Output: 12bit digital output • Data Acquisition Unit (DAU) Specification - Power Supply: 5V/ 200mA - Data Output: USB 3.0 - Data Input: USB 3.0 - Data Synchronization: 10ns accuracy - Withstand Voltage: 500V isolation. - Output: 12bit Digital Output
R&D Plan	**[R&D Flow]** A. Quantification of commercial product (SME) • Target equipment: Large electric motors, transformer energization, power reactors (saturated), capacitor banks, etc. • Installation constraints: Non-invasive (Hall clamp), outside the cubicle, optical/Ethernet, safety/insulation • Output definition: Real-time R/L/C + history graph + diagnostic indicators (alarms/trends) • Definition of success criteria (KPIs) • Examples: R/L estimation error (normal) within ±5%, maintenance of saturation region trend, latency < 0.5 s, etc. • Deliverables: System Requirements Specification (SRS), installation/operation procedures, KPI/test item list B. Develop Equivalent Model and Algorithm (SME/Purdue) • Definition of equivalent circuits (frequency-dependent) for each motor/reactor/transformer • Modeling to determine "what minimum frequency components are needed to separate RLC components" • Definition of observable conditions (e.g., if V3 is too small, C is unstable → C is treated as a constant, etc.) • Deliverables: Estimable operating envelope for each equipment, inestimable conditions/corresponding strategies • Signal acquisition/preprocessing/synchronization (measurement architecture) • Confirmation of Hall CT (3-phase) + PT input structure

- Synchronization: Voltage/current timestamp alignment, phase error correction
- Preprocessing: anti-aliasing, DC removal, windowing (Hann), frequency estimation (PLL)
- Communication: Sensor main unit (Ethernet/optical) protocol, handling time delay/packet loss
- Deliverables: Data acquisition board/firmware, sampling and synchronization design, communication specifications
- Develop RLC Extraction Algorithm (Purdue)

- Stabilization when the 3rd/5th harmonic components are small (weighting, multiple window averaging, C constant setting)
- Frequency deviation (59.8~60.2Hz) handling (using accurate ω based on PLL)
- Output: Algorithm library (Python/MATLAB reference + C/C++ porting design)
- CT/VT saturation handling (including correction/restoration, not just skipping)
- Saturation detection: I2/I1, sudden phase change, waveform asymmetry index
- CT flux restoration requires "field CT parameters," therefore:
- Burden (Ω), knee point, and magnetization curve parameters are obtained through testing/field calibration.
- Output: Saturation handling logic, restoration parameter calculation procedure, test report template

C. Develop Real-time Online Device (Embedded) (SME)

- Real-time constraints: Computation cycle (e.g., 50-200 ms update), CPU/memory budget
- Implementation priorities:
  - Primary + tertiary based R/L first (stable)
  - Add fifth order for improved accuracy
  - Option to mark C as "trend/reference" if reliability is low
  - UI/UX: Real-time R/L/C + History graph + event markers
- Deliverables: Firmware/software, HMI screen, data logging/reports

D. Validation (SME/Purdue)

3 stage validation: Lab → Pilot → Plant deployment

[Key Tasks and Job Partitioning]

- Define detailed specifications and application/commercialization scenarios (Korean SME)
- Define a physics-based RLC model and typical model parameters (Korean SME & Purdue)
- Develop sensing hardware and software design (Purdue)
- Develop RLC parameter extraction algorithm (Purdue)

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	• Implement and customize the sensing hardware and software (Korean SME)
	• Deployment of hardware in a nuclear power plant and data collection (Korean SME)
	• Develop a time-series-data-based diagnostic algorithm (Purdue)
	• Field evaluation of the diagnostic algorithm (Korean SME)
	
	[Timeline and Milestones]
	
	• Phase 1 (Year 1) – Validate core technology
	  ▪ Design a non-intrusive current and voltage sensing system
	  ▪ Develop a high-precision amplitude and phase extraction algorithm
	  ▪ Develop a harmonic extraction algorithm
	  ▪ Develop an RLC model and a parameter extraction algorithm
	  ▪ Set up lab test environment
	• Phase 2 (Year 2) – Hardware and Algorithm Implementation & Deployment
	  ▪ Optimization of the non-intrusive voltage and current sensing hardware
	  ▪ Implementation of online monitoring and extraction algorithm on the edge
	  ▪ Implementation of a prototype, including hardware and software
	  ▪ Pilot test at a nuclear power plant
	  ▪ Development of a time-series-based diagnosis algorithm
	• Phase 3 (Year 3) – Data collection and implementation of the diagnosis algorithm
	  ▪ Long-term deployment and data collection
	  ▪ Implementation of a time-series-based diagnosis algorithm
	  ▪ Develop a diagnosis guideline
	  ▪ Preparation for the standardization of the diagnosis method

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Development of a magnetic field-sensing-based current sensing and monitoring unit • Development of a minimally intrusive voltage sensing and monitoring unit • Development of a voltage and current-based R-L-C extraction algorithm and its implementation on an edge computing unit • Development of communication links between the sensing units and the data acquisition unit • Development of communication links between the data acquisition unit • and the external systems • Development of an RLC data-based diagnostic algorithm
Expected R&R of Korean SME	• Implementation of AI-based diagnostic algorithms • Hardware prototype manufacturing

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Partner	• Develop a remote load monitoring and diagnostics system • System design and construction • On-site installation and testing, field data collection • On-site operation and verification • Certification, mass production, and commercialization planning

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Reduction in accidents at nuclear power plants and substation power facilities • Establishment of a power facility fault prevention and diagnosis system • Transition from TBM (Time-Based Maintenance) to CBM (Condition-Based Maintenance) for cost-efficient equipment management • Significant reduction in equipment inspection time and maintenance costs • Improved power system reliability and accident prevention • Integration into KEPCO's preventive diagnosis system to improve system reliability and build a remote inspection infrastructure • Creation of a high-value-added global market for next-generation intelligent power grid • Potential for expansion into preventive diagnostics systems for industrial facilities, such as semiconductor, automotive, and steel factories.